

THE INTERCONNECTION QUEUE, NOT THE CHIP, IS THE BINDING CONSTRAINT ON AI COMPUTE

Technical write-up | Data centre infrastructure | Independent research
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The trend	Over the last twelve months the gating item on data centre buildout has moved from generation capacity to grid connection. Power is available. Wires, substations, transformers and queue position are not.
The claim	Announced megawatts and energised megawatts are diverging. The scarce asset is a completed connection, and it is not priced as scarce.

The constraint moved from megawatts to wires, and two markets show it clearly

I care about this because it is the one part of the AI buildout where physics sets the clock and the constraint is legible in public data rather than inferred from supply chain checks. Johor is the cleanest example. Data centre load there more than doubled between 2024 and 2025 and reached roughly 3.8GW of maximum demand, which Wood Mackenzie notes is about 1.5 times the state's own current electricity demand. Generation capacity remains sufficient system wide. Access to transmission and distribution is what now decides whether a project proceeds. Inside the Johor-Singapore Special Economic Zone, installed capacity stood at 3,885 MVA against demand near 1,272MW in December 2025, a 72.76% utilisation rate that is comfortable today and narrows quickly at the growth rate of the last two years.

The utility's own behaviour is the strongest evidence. Tenaga Nasional has committed RM43bn, around US\$10.8bn, to modernise the grid specifically for data centre demand, and its Green Lane Pathway cut new connection timelines from 36 months to 12, delivering 33 projects under that framework by March 2026. You do not build a programme to compress connection time unless connection time, not generation, was the thing stopping projects. Singapore tells the same story from the other side: peak demand is projected to grow 3.7-5.7% a year to 10.1-11.8GW by 2030, and the response has included a roughly 20 hectare low carbon data centre park on Jurong Island sized to 700MW and a capacity call requiring a PUE of 1.25 or better. Singapore is rationing connection by efficiency because it cannot ration it by supply.

Why the market still prices this as a semiconductor problem

A GPU order clears in months. A substation, its transformer and its queue slot clear in years, and the transformer is itself supply constrained. That asymmetry means announced capacity has become a poor proxy for deliverable capacity, and the gap shows up late and unglamorously, as slipped commissioning dates and slipped revenue recognition at operators. Accelerator supply is tracked weekly by a large sell side community; interconnection queues are tracked by very few people with capital at risk. That is where I think the mispricing sits, and it is why a power and grid read is worth something next to a silicon read rather than instead of one.

Where the value accrues if the claim is right

First, the equipment that clears the queue: high voltage transformers, cable, switchgear and substation EPC, where order books extend past the capex cycle that funds them. Second, operators holding energised or queue secured capacity, which should trade at a premium to peers quoting announced megawatts, a distinction disclosure rarely makes cleanly. Third, storage, which is quietly becoming an interconnection product rather than only an arbitrage one. US utility scale battery additions are expected at 24GW in 2026 against 15GW in 2025, Texas alone around 12.9GW, more than half the national total, sited against ERCOT data centre load that ERCOT itself projects near 35GW of peak demand by 2035. A battery that lets a campus energise ahead of its wires is worth more than its arbitrage spread.

What would prove me wrong, and what I am watching

The honest falsifier is Tenaga's own success. If connection timelines keep compressing the way the Green Lane did, from 36 months to 12, the premium on queue position decays fast and this becomes a two year dislocation rather than a structural one. I would also be wrong if transformer and high voltage cable lead times normalise faster than utility capex is committed, which would turn a bottleneck into a glut. So I watch four things: transformer and HV cable lead times, Green Lane project count beyond 33, JS-SEZ utilisation past 72.76%, and the composition of the Singapore capacity award, because who wins 200MW under a 1.25 PUE ceiling tells you what the regulator now believes is buildable.

Sources: Energy Market Authority, EDB and JTC, Tenaga Nasional disclosures, Wood Mackenzie, EIA and ERCOT, drawn from my own market notes on the Singapore-Johor power corridor (29 July 2026) and grid scale battery storage (7 August 2026). Figures are as reported on the dates cited. Personal research, not investment advice.